

Calcium containing feedstock processing

Development of a FCC catalyst/additive combination with high tolerance to calcium contamination from lower cost feedstock

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Refiners continually find ways to improve refining margin by processing cheaper feedstocks, such as heavy oil, resid and opportunity crudes in their FCC units. Whenever processing of resid or some cheaper feedstock is increased in the FCC unit, plant operating severity is adjusted. Both metals and basic nitrogen compounds, which are known to poison FCC catalysts, are concentrated in the heavier end of gas oils and resids. These poisons, mostly present in the heavier hydrocarbon molecules, deposit on the catalyst during cracking reactions.

Each type of poison affects the FCC catalyst differently. Refiners are conversant with the detrimental effects of vanadium, basic nitrogen, nickel and their treatments. Vanadium deactivates Y-zeolite by the formation of vanadic acid in the presence of steam at high temperature and sodium facilitates the dealumination reaction by forming a low eutectic with the rare earth metals present in Y zeolite. Nickel promotes dehydrogenation reactions

leading to high fuel gas make and coke formation and hence reduces the selectivity of desired products. Iron and calcium metals deposit on the catalyst surface and cause a loss of diffusivity, which leads to a loss in conversion and an increase in coke and bottoms. Deposition of iron on FCC catalyst reduces accessibility to the catalyst pore and consequently reduces the catalyst's activity. Researchers have studied the effect of calcium and iron on coke formation over ultra-stable Y-zeolite catalyst in the absence and presence of nickel and vanadium metal.¹ Different zeolite samples are prepared by impregnating nickel and vanadium on ultra-stable Y-zeolite, previously exchanged with calcium. The catalyst samples are used for cracking n-hexane at 500°C. The study showed that catalyst containing calcium in combination with nickel and vanadium reduces coke formation significantly and increases the olefin to paraffin ratio.

There is plenty of reported information on the effects of contaminant nickel, vanadium, sodium and other metals in the

FCC.² Guthrie *et al* described passivating the reactivity of contaminant metals, such as nickel and vanadium, which deposited on a catalytic cracking catalyst, by adding to a cracking catalyst a mixture of a calcium containing material and a magnesium containing material in a separate reactor in the presence of steam.² The preferred calcium containing material was dolomite and the preferred magnesium containing material was sepiolite. The cheaper feedstock contains metal contaminants, which contribute to lower conversion and the production of more fuel gas and coke. The higher fuel gas yield often touches the reactor cyclone velocity limits, which results in lower severity operation in the FCC unit, such as lower riser temperature. Similarly, higher coke yield leads to a higher regenerator temperature that lowers unit conversion.

However, there have not been many studies focused specifically on calcium contaminants and their effect on the performance of FCC catalysts/additives. Some crude samples

Typical compositions of catalyst and ZSM-5 additive formulations

Additive/catalyst composition	Catalyst	Additive
USY, wt%	33	0
ZSM-5, wt%	0	45
P ₂ O ₅ , wt%	0	11
Alumina, wt%	12	5
Silica sol, wt%	23	23
REO, wt%	0.5	0
	Balance clay	Balance clay

Table 1

have a higher concentration of calcium, alone or with other conventional contaminant metals (Na, Ni and V). Hence, a study was undertaken on the effects of calcium on FCC catalyst/additive on their performance in a fixed fluidised bed reactor. The catalyst and additive were optimised with high zeolite and matrix components to resist the effects of contaminants like calcium in the feed. The performance results are correlated with physico-chemical characterisations to gain a better understandings of the effect of calcium on catalyst and additive.

Experimental

Catalyst preparation

FCC catalyst and additive were prepared according to procedures mentioned in earlier patent disclosures.³ ZSM-5 additive was prepared by mixing the required quantities of kaolin clay, boehmite alumina, phosphate salt, ZSM-5 zeolite and colloidal silica with a suitable dispersant to obtain a free flowing slurry, which was then subjected to spray drying to form catalyst microspheres. Two kinds of phosphorus source, H₃PO₄ and monoam-

monium hydrogen phosphate were used to introduce phosphorus to the ZSM-5 additive formulation. The composition details of the catalyst and additive are shown in **Table 1**. The obtained spray dried microsphere particles were calcined at 500°C for one hour prior to hydrothermal deactivation.

The FCC catalyst was prepared by mixing the required quantities of kaolin clay, peptised boehmite alumina, USY-zeolite and colloidal silica with a suitable dispersant to obtain a free flowing slurry, which was then subjected to spray drying to form catalyst microspheres. The spray dried catalyst was exchanged with lanthanum nitrate (a rare earth salt) and then the sample was calcined at 500°C for one hour prior to hydrothermal deactivation.

A modified Mitchell method⁴ was followed for the impregnation of calcium on catalyst and additive separately, followed by calcination at 590°C for three hours. The calcium content varied from 0 ppm to 10 000 ppm using calcium naphthenate as the source of calcium precursor since calcium is present in crude mostly in the form of calcium naphthenate.

All calcium doped catalyst and additive samples were then hydrothermally deactivated at 800°C for 20 hours using 100% steam at atmospheric pressure before the cracking reaction took place.

BET surface area and pore volume measurements

Nitrogen gas adsorption/desorption isotherms were obtained using a Micromeritics ASAP 2020 unit. The catalysts were degassed for two hours at 300°C prior to adsorption. Nitrogen gas was dosed very precisely for both adsorption and desorption processes to generate highly accurate isotherm data. The BET surface area was determined by considering relative pressure (P/P₀) between 0.05 to 0.20 and pore volume at 0.98 relative pressure. The t-plot was used to calculate the external surface area of the catalyst particles (calculated as the surface area of pores larger than micropores).

NH₃-TPD measurements

Ammonia temperature programmed desorption (TPD) experiments were carried out on a Micromeritics Autochem 2920 unit equipped with a thermal conductivity detector (TCD). The sample was pre-treated at 600°C under a flow of helium gas for an hour. The sample was saturated with ammonia at 120°C for 30 minutes and ammonia was flushed out subsequently at the same temperature in a helium flow for an hour to remove weakly adsorbed ammonia. TPD analysis was carried out from 100°C to 600°C at a heating rate of 10°C/min. The

desorbed ammonia was quantified with TCD and the signal was plotted against time/temperature.

Cracking reaction

The advanced cracking evaluation (ACE) unit is a fixed fluid bed reactor for the evaluation of catalysts and additives, feedstocks and process development. It is downflow with respect to feed injection, an isothermal tubular reactor equipped with a central thermowell to measure temperature in the catalyst bed. The reactor is heated by an electric furnace with a minimum of three separate heating sections, which allow fine control for isothermal operation. It includes control system hardware and software that enables accurate multiple runs without operator intervention.⁵ Cracking reactions in the ACE reactor take place in conditions that simulate a commercial FCC riser (see **Table 2**). The conversion is varied by changing the catalyst loading at constant feed rate.

The calcium impregnated catalyst and additive were hydrothermally deactivated at 800°C for 20 hours prior to performance evaluation. The catalyst and additive are used as a mixture of 75% catalyst and 25% additive for evaluation. Reaction was carried out at four different catalyst to oil ratios (from 4 to 10) to generate a wide range of conversion data. The reaction temperature was maintained at 545°C. Product gas and liquid were analysed in an Agilent 3000A micro GC and a Varian 450 GC SIMDIST respectively.

ACE operating conditions

Parameter	ACE protocol
Feed injection time	Fixed at 30 seconds
C/O range	4 to 10
Feed rate, gm/min	2.0
Rx temp., °C	545
Rx. pressure, kg/cm ² (g)	Atmospheric operation

Table 2

Results and discussion

Physico-chemical properties of the FCC catalyst

The spray dried catalyst after rare earth exchange and calcination is analysed by various physico-chemical techniques and the results are summarised in **Table 3**.

The total surface area of prepared FCC catalyst is 336 m²/g, which decreases to 165 m²/g after hydrothermal deactivation at 800°C for 20 hours (see **Tables 3** and **4**). The decrease in surface area of steamed FCC catalyst is mainly due to dealumination of the zeolite, resulting in a loss of Al-OH-Si groups responsible for Bronsted acidity; hydrothermal treatment leads to partial destruction of the Y-zeolite structure and thereby a significant decrease in surface area, pore volume and acidity

Physico-chemical properties of prepared FCC catalyst (without calcium impregnation)

Surface area and pore volume	
Total surface area	336 m ² /g
Zeolite surface area	226 m ² /g
Matrix surface area	110 m ² /g
Total pore volume	0.203 cc/g
Zeolite pore volume	0.049 cc/g
Chemical analysis and acidity	
Al ₂ O ₃ , wt%	29.4
Na ₂ O, wt%	0.28
RE ₂ O ₃ , wt%	0.54
Total acidity, mmol/g	706
Average particle size distribution and attrition index	
APS, µ	71
Fines (<40 µ), %	4
Attrition Index (ASTM D5757) (wt% loss in 5 hrs)	3

Table 3

is anticipated. The decrease further depends on the severity of the hydrothermal deactivation conditions and stability of zeolite, which is formulation specific. Exchange of rare earth retards destruction of the Y-zeolite during hydrothermal treatment, which also results in an increase in the strength of the acid sites. The average particle size (APS) and attrition index (AI) results (see **Table 3**) are within the range of required specifications of commercial catalyst.

Acidity measurements were

Surface area, pore volume and acidity of hydrothermally deactivated FCC catalyst samples having calcium contents from 0 to 1 wt%

Parameters	FCC catalyst (hydrothermally deactivated)		
Calcium concentration on catalyst, ppm	0	5000	10 000
Total surface area, m ² /g	165	149	137
Zeolite surface area, m ² /g	107	107	92
Matrix surface area, m ² /g	58	42	45
Total pore volume, cc/g	0.203	0.187	0.180
Zeolite pore volume, cc/g	0.049	0.049	0.042
Total pore volume reduction, %	Base	7.9	11.3
Total acidity, mmol/g	0.044	0.044	0.036
Acidity reduction, mmol/g	Base	Nil	18

Table 4

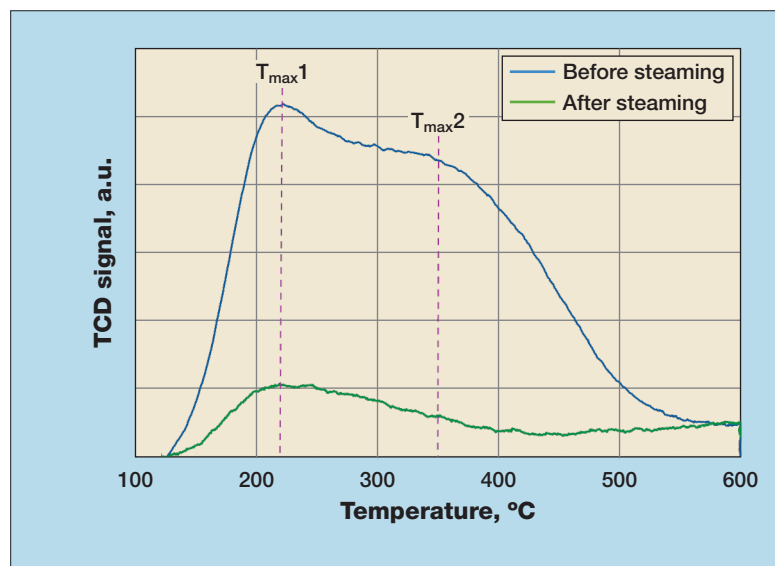


Figure 1 Ammonia TPD profile of FCC catalyst before and after hydrothermal deactivation

carried out by the ammonia TPD method. The ammonia TPD profile of prepared FCC catalyst before and after hydrothermal deactivation is shown in **Figure 1**. It can be seen from **Figure 1** that the acidic sites are distributed in two regions: one is due to weak acidic sites ($T_{\max 1}$: $\sim 230^{\circ}\text{C}$) and the other is due to strong acidic sites ($T_{\max 2}$: $\sim 350^{\circ}\text{C}$). The total acidity of catalyst prior to hydrothermal deactivation is about 0.706 mmol/g, decreasing to 0.044 mmol/g after steaming. The decrease in acidity (both weak and strong acidic sites) is very rapid in FCC catalyst and the amount of acid sites present in the catalyst are minimal (0.044 mmol/g) after hydrothermal deactivation.

The calcium impregnated (0, 5000 and 10 000 ppm) FCC catalyst samples are hydrothermally deactivated and their physical properties are shown in **Table 4**. The total surface area of calcium impregnated catalyst samples reduces to 137

m^2/gm from 165 m^2/gm with an increase in calcium level from 0 to 10 000 ppm. The decrease in total pore volumes are also similar. The question is whether the decrease in these properties is due to zeolite pore blockage or blockage of matrix pores (mesopores) by calcium. To understand further, a t-plot was used to

Physico-chemical properties of prepared ZSM-5 additive (without calcium impregnation)

Surface area and pore volume	
Total surface area	140 m^2/g
Zeolite surface area	110 m^2/g
Matrix surface area	30 m^2/g
Total pore volume (cc/g)	0.134
Zeolite pore volume (cc/g)	0.053
Chemical analysis and acidity	
Al_2O_3 , wt%	18.7
Na_2O , wt%	0.11
P_2O_5 , wt%	11.9
Total acidity, mmol/g	0.240
Average particle size distribution and attrition index	
APS, μ	77
Fines (<40 μ), %	6
Attrition Index (ASTM D5757) (wt% loss in 5 hrs)	4

Table 5

calculate the matrix surface area (pores larger than micropores) and acidity (ammonia TPD) of the catalyst particles.

It is interesting to note that the zeolite surface area (107 m^2/g), zeolite pore volume (0.049 cc/g) and total acidity (0.044 mmol/g) are the same and did not change when calcium is present in the catalyst up to 5000 ppm (see **Table 4**). The decrease in matrix surface area from 58 m^2/g to 42 m^2/g with no change in zeolite surface area suggests that impregnated calcium is mostly deposited in the catalyst matrix and possibly has not blocked the zeolite pores. However, when the calcium content is increased to 10 000 ppm, then the calcium partially filled both zeolite and matrix pores, revealed in the decrease in zeolite surface area, zeolite pore volume and acidity. Hence the prepared FCC catalyst is capable of capturing calcium up to 5000 ppm. **Figure 2** shows the NH_3 -TPD profiles of hydrothermally deactivated catalysts with calcium contents from 0 to 10 000 ppm. Similar to the surface area results, the total acidity (weak acid sites) is also comparable, or not much changed, when the calcium content is 0 ppm to 5000 ppm; acidity drops thereafter as zeolite surface area is affected by an increasing calcium level at 10 000 ppm.

Physico-chemical properties of ZSM-5 additive

The spray dried catalyst additive after calcination is analysed and various physico-chemical properties are summarised in **Table 5**.

The total surface area of

prepared ZSM-5 additive is 140 m²/g which increases to 170 m²/g after hydrothermal deactivation at 800°C for 20 hours (see **Tables 5** and **6**). The behaviour of steamed ZSM-5 additive is different to that of FCC catalyst containing Y-zeolite. In contrast to FCC catalyst, steaming of ZSM-5 additive results in either an increase in total surface area or the same total surface area, depending on the relocation of aluminum/phosphate sites in the zeolite framework and matrix. The surface area also depends on the severity of hydrothermal deactivation conditions and the stability of the zeolite. It is well known in prior art that the hydrothermal stability of ZSM-5 zeolites in FCC additive is stabilised by phosphates. Although dealumination occurs in ZSM-5 zeolite, its crystal structure is quite stable under hydrothermal deactivations, hence not much crystallinity and surface area loss is observed. In fact, the surface area increases, particularly that of the matrix in ZSM-5 additives after hydrothermal deactivation. This is due to relocation of aluminum and phosphorous sites outside the framework. The zeolite surface area also decreases if the sample undergoes severe hydrothermal deactivations, leading to dealumination. The zeolite structure too collapses and this can be seen in a lower zeolite surface area and crystallinity. The surface area further depends on the phosphate contents and binders used in the formulation. In the prepared ZSM-5 additive, the zeolite surface area decreased to 93 m²/g from 110 m²/g and

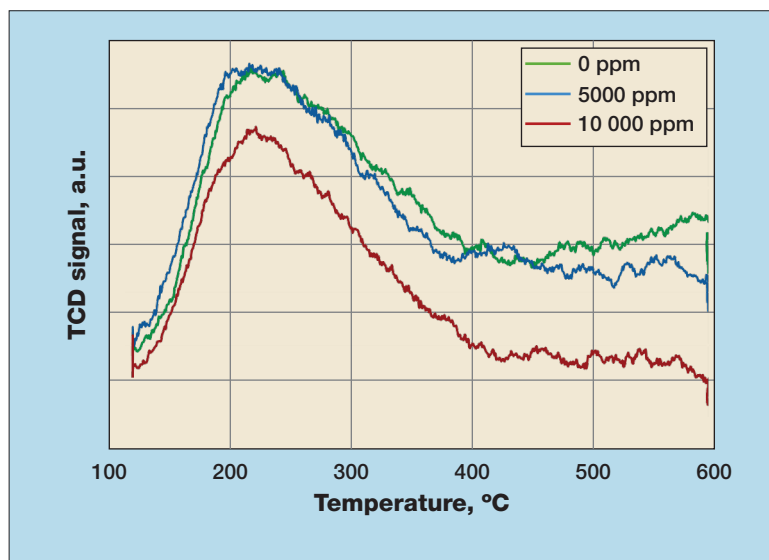


Figure 2 Ammonia TPD profile of hydrothermally deactivated catalyst with calcium contents from 0 to 10 000 ppm

the matrix surface area increased to 77 m²/g from 30 m²/g. The average particle size and attrition index results are within the range of desired specifications.

Figure 3 shows the ammonia TPD profile of prepared ZSM-5 additive before and after hydrothermal deactivation. The total acidity of ZSM-5 additive is observed to be 0.240 mmol/g, which decreases to 0.100 mmol/g. Unlike FCC catalyst, the decrease in acidity of ZSM-5 additive is not significant. The total acid sites

present in steamed additive are higher (0.100 mmol/g) in comparison to steamed catalyst (0.044 mmol/g). The acid sites present in fresh additive are mostly weak acids with some strong acid sites. However, only weak acid sites are present in both catalyst and additive after steaming.

The calcium impregnated (0, 5000 and 10 000 ppm) ZSM-5 additive samples are hydrothermally deactivated and their physical properties are shown in **Table 6**. The results show that the physical properties of

Surface area, pore volume and acidity of hydrothermally deactivated ZSM-5 additive samples with calcium content of 0-1 wt%

Parameters	ZSM-5 additive (hydrothermally deactivated)		
Calcium concentration on additive, ppm	0	5000	10 000
Total surface area, m ² /g	170	166	148
Zeolite surface area, m ² /g	93	88	77
Matrix surface area, m ² /g	77	78	61
Total pore volume, cc/g	0.153	0.151	0.136
Zeolite pore volume, cc/g	0.042	0.040	0.035
Total pore volume reduction, %	Base	1.3	11.1
Total acidity, mmol/g	0.100	0.090	0.062
Acidity reduction, mmol/g	Base	10	38

Table 6

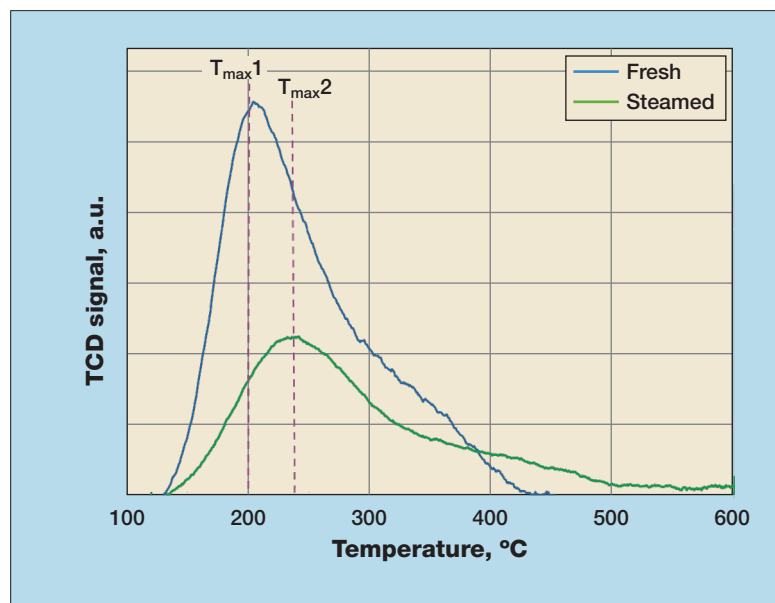


Figure 3 Ammonia TPD profile of ZSM-5 additive before and after hydrothermal deactivation

the FCC additive are not changed appreciably when the calcium concentration is between 0.0 ppm and 5000 ppm. A marginal decrease in zeolite surface area and acidity are observed in 5000 ppm calcium containing additive. This indicates calcium is

partially filling zeolite pores at lower concentration levels (<5000 ppm). However, at higher concentrations (>5000 ppm), calcium fills in both the zeolite and matrix pores of the ZSM-5 additive. The total surface area reduced from 166 m²/g to 148 m²/g due to a

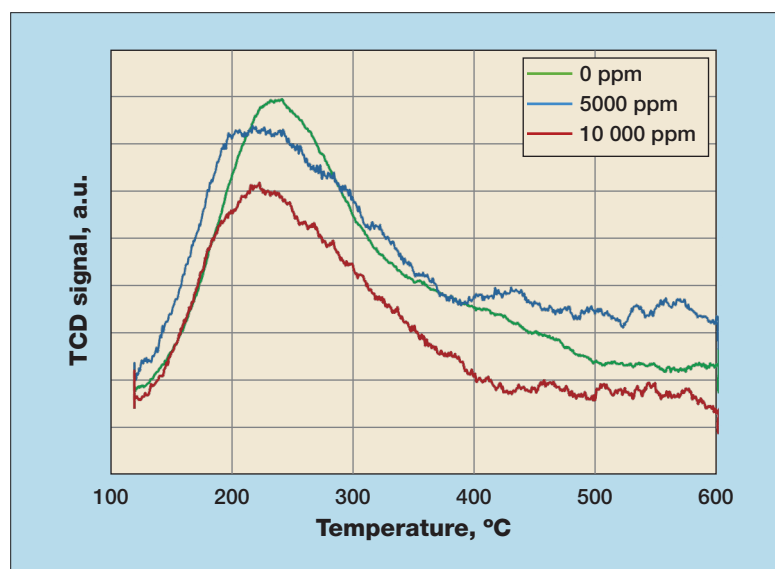


Figure 4 Ammonia TPD profile of hydrothermally deactivated ZSM-5 additive with calcium content from 0 to 10 000 ppm

decrease in zeolite surface area, as well as matrix surface area in a 10000 ppm calcium containing additive. Similarly, pore volumes and acidities decrease significantly when the calcium level is 10 000 ppm.

Figure 4 shows the NH₃-TPD profiles of hydrothermally deactivated ZSM-5 additives with calcium contents from 0 ppm to 10 000 ppm. Similar to the surface area results, the acidity is also marginal decrease in 5000 ppm Ca and the fall in acidity is much higher when the calcium content is 10 000 ppm.

Fluid catalytic cracking of hydrotreated VGO on calcium impregnated catalyst/additive

Calcium impregnated, steam deactivated catalyst and additive were evaluated in the ACE at 545°C using hydrotreated VGO feed. Conversion at a C/O ratio of 8.2 and product selectivities at a conversion level of 76 wt% are shown in **Figures 5 to 8**. Conversion vs calcium concentration is plotted in **Figure 5**, which shows that the catalyst has lost its activity by 1 wt% and 1.7 wt% when the calcium level was increased to 5000 ppm and 10 000 ppm, respectively. Escobar *et al* reported monometallic catalysts containing iron or calcium were less active than a USY-zeolite sample.¹ Both iron and calcium favour the olefin to paraffin ratio compared to the metal free sample. These metals increase olefin formation and poison acid sites. The effect of calcium concentration on fuel gas, coke and CSO yields are shown in **Figure 6**. The results show the decrease in fuel gas yield with increase in calcium

level. It is interesting to note that fuel gas make reduces to 2.9 wt% from a base level of 4.75 wt% when the calcium level on catalyst/additive increases to 10 000 ppm. The coke selectivity plot shows that coke yield increases with an increase in calcium level and the increment is significant at 10 000 ppm. These findings contrast with the nickel effect. Nickel makes more coke and more fuel gas, whereas calcium also shows more coke but reduces fuel gas considerably. In the current study, the decrease in fuel gas yield can be attributed to lower mono-molecular and bimolecular reactions, which is also related to a decrease in acid sites. Propylene and LPG yields were also lower in 10 000 ppm calcium containing samples.

The effect of calcium on propylene yield is shown in **Figure 7**. The plot shows that up to 5000 ppm calcium level, there was not much change in propylene yield, but an appreciable decrease in propylene yield was observed when the calcium concentration was increased to 10 000 ppm. **Figure 8** shows the effect of calcium on the yield of LPG, gasoline and LCO. The decrease in LPG and increase in gasoline yields at 10 000 ppm calcium level could be attributed to reduced catalyst activity of the additive.

Estimated heat balanced yields

For prediction of heat balanced plant yield based on ACE data, conversion and coke factors are fed to the FCC models to obtain heat balanced conversion for the calcium doped catalyst/additive. Selectivity deltas, thereafter, are imposed

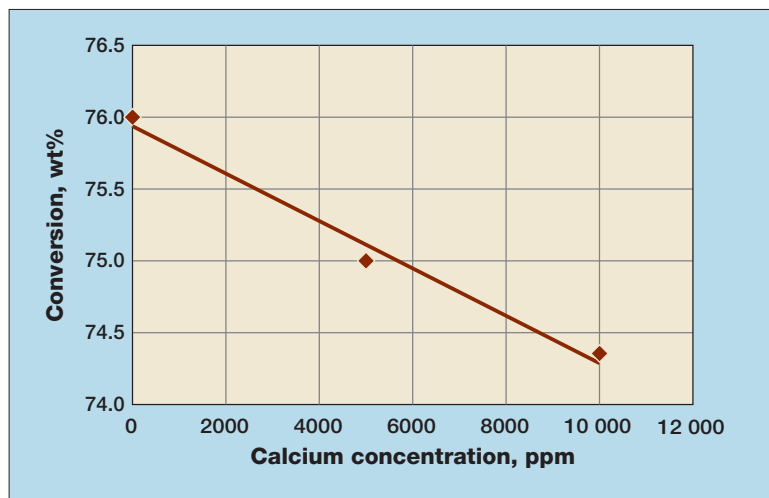


Figure 5 Conversion vs calcium loading from 0 to 10 000 ppm in FCC catalyst and additive

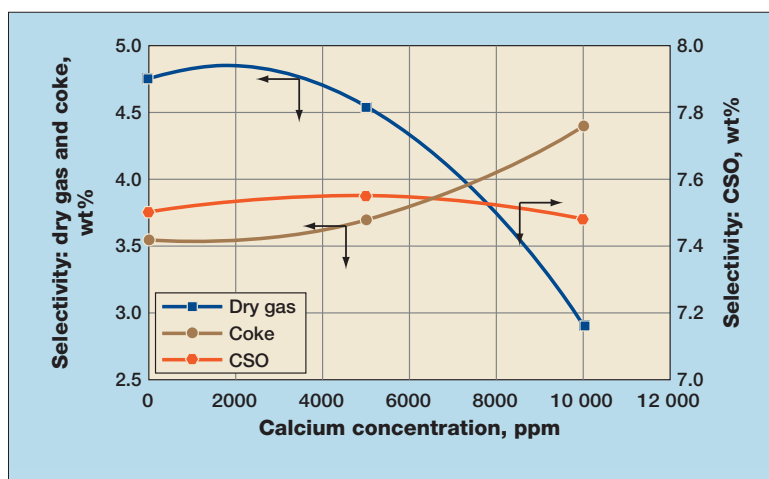


Figure 6 Effect of calcium concentration on dry gas, coke and CSO yields in the FCC

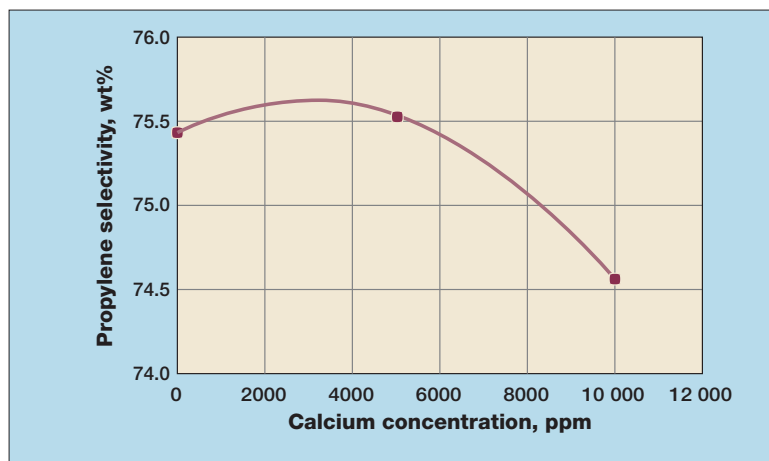


Figure 7 Effect of calcium concentration on propylene yield in the FCC

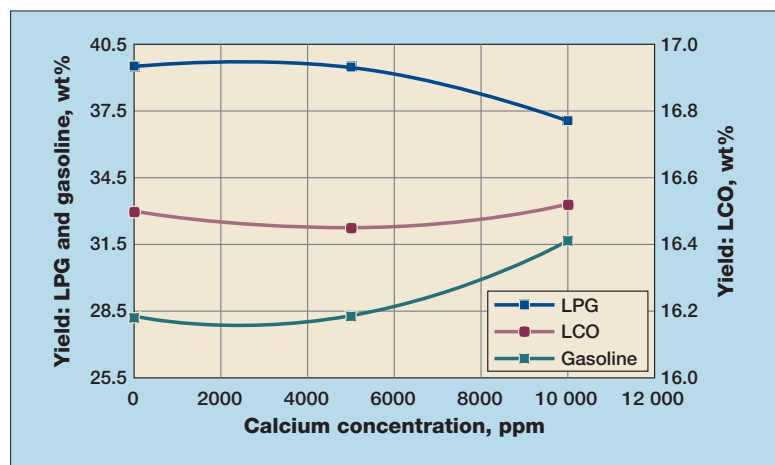


Figure 8 Effect of calcium concentration on yields of LPG, gasoline and LCO in the FCC

at the new conversion to predict final yield shifts with the calcium impregnated catalyst/additive. The estimated heat balanced yields are shown in **Table 7**. It can be concluded that the catalyst loses its conversion by 1.2 wt% and 2.1 wt% when the calcium level on the catalyst was increased to 5000 ppm and 10 000 ppm respectively.

K Bryden *et al* studied the detailed characterisation of tight oils and cracking of these feed-

stocks under different operating conditions.⁶ Tight oils are generally light, sweet and easy to crack and contain sediments with high levels of iron and alkali metals. Iron in combination with calcium and/or sodium has a stronger negative effect on catalyst performance than iron alone. It is reported that iron and calcium poisoning results in a loss of bottom cracking due to pore blockage, leading to a drop in conversion. However, the calcium tolerance

of these catalysts for FCC feedstocks in general are not reported.

The fuel gas make reduces to half of the base level as the calcium level on catalyst increases to 10 000 ppm. LPG as well as propylene selectivity for the base case and the 5000 ppm calcium case is almost the same. However, when the calcium level increases to 10 000 ppm, the drop in LPG and propylene is significant. As per the heat balance yield, LPG and propylene make drop by 3.2% and 1.18% at the 10 000 ppm calcium level. Gasoline make increases to 43.7% from a base value of 39.9 wt%. The secondary cracking of gasoline to lighter products including LPG and propylene is significantly reduced at the 10 000 ppm calcium level. The change in LCO and CSO make is not much at 5000 ppm calcium. However, LCO yield with 10 000 ppm calcium, increases to 14.2% from a base value of 13.4%. Similarly, CSO make increases to 8.12 wt% from 7.38% in the presence of 10 000 ppm calcium.

Estimated heat balanced yields at constant ROT of 545°C

Parameters	Base case		Ca doped case	
	0	5000	10 000	
Calcium , ppm	0	5000	10 000	
Yield pattern, wt%				
Fuel gas	4.09	3.59	2.0	
Total LPG	29.84	29.88	26.60	
LPG (excluding C ₃ =)	20.12	20.09	18.06	
Propylene	9.72	9.79	8.54	
Gasoline	39.89	39.70	43.74	
LCO	13.37	13.74	14.20	
CSO	7.38	7.69	8.12	
Coke	5.44	5.41	5.32	
Conversion	79.8	78.6	77.7	
Process conditions				
Riser outlet temp, °C	545	545	545	
Reactor pressure, kg/cm ² (g)	2.82	2.82	2.82	
Cat/oil	10.28	10.25	9.96	

Table 7

Conclusions

The catalyst and additive in this study are tolerant to calcium metal contamination up to 5000 ppm without affecting the product yields of propylene and LPG. The catalysts lower the additional fuel gas by 12% and 50% when the calcium concentration is 5000 ppm and 10 000 ppm respectively. Coke yields are observed to increase with calcium level and the increment is significant at 10 000 ppm calcium. The calcium effect is similar to the nickel

effect with respect to coke make; however, it contrasts with nickel with respect to fuel gas. Therefore the catalyst and additive provide the feasibility for processing inferior quality hydrocarbon feedstock of higher boiling point. The catalyst was found to lose activity by 1.2% and 2.1 wt% when the calcium level on the catalyst increases to 5000 ppm and 10 000 ppm respectively. Further, LPG and propylene yields were found to decrease considerably at the 10 000 ppm calcium level and gasoline make increased to 43.7% from a base value of 39.9 wt%.

Surface area and pore volume results show that initial pore blockage is not much in the catalyst and additive up to the 5000 ppm calcium level. At higher calcium loading (10 000 ppm), both zeolite and matrix pores are affected in the catalyst and additive, with a decrease in surface area and pore volume. This is due to calcium filling zeolite as well as matrix pores in the catalyst and additive at higher concentrations (>5000 ppm).

Acidity measurements further confirm the observations of surface area and pore volume. The decrease in acidity of catalyst and additive was marginal or the same up to the 5000 ppm calcium level and acidity decreased when calcium was >5000 ppm. Therefore, it can be concluded that up to 5000 ppm calcium on catalyst/additive, there was negligible effect on

desirable yields. The reduction in activity and product selectivities could be explained due to an appreciable amount of reduced acid sites in the ZSM-5 additive and catalyst when calcium concentration was >5000 ppm. Hence, the catalyst and additive of the current study has a potential to process calcium containing feedstock in the FCC. These catalysts may also be employed in FCC units where fuel gas is the major constraint.

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